LESSONS LEARNED REVIEW

Fireline Chainsaw Operations, Chainsaw Fuel Geyser, Burn Injury

July 29, 2016

Pioneer Fire – Boise National Forest, Idaho



What is left of the saw after it was "put out" with dirt by the crew.

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1. Summary

Chainsaw Fuel Geyser Results in 2nd Degree Burns
On Friday, July 29, at approximately 1400, a sawyer received 2nd degree burns to his lower arm when fuel from his chainsaw sprayed onto his arm and his saw and then combusted.

The sawyer was part of a two-man felling team, working ahead of the crew, falling hazard trees. When the chainsaw *"seemed like it was out of fuel"* the sawyer walked away from the hot area in which he was working and opened the cap to confirm. That's when fuel "geysered" from the saw and sprayed the sawyer and his chainsaw. The sawyer's quick reaction of using his gloved hand to block the geyser protected his face, but directed the fuel along the bottom of his arm and inside the sleeve of his Nomex shirt.

"A fuel geyser is not the little hiss and spray that we are used to. All the hotshot crews know and talk about the geysers, but this is more than we understand it to be."

Assistant Crew Superintendent

The fuel ignited, engulfing the saw and the sawyer's arm. The sawyer quickly tossed the saw away when he realized it was on fire. At that moment, he was unaware that he himself was on fire until his swamper pointed it out to him. He then got to a green area, shed his pack, and rolled to extinguish the flames.

The Crew EMT responded immediately, providing initial assessment and treatment. It was determined that the injured sawyer was able to hike off the fireline under his own power and meet the Division Paramedic at the road.

Within an hour of the first report, the injured sawyer was transported to a helispot by the Paramedic in his SUV. Once at the helispot the injured sawyer was flown to the helibase by the Pioneer Fire's identified medevac helicopter and then loaded onto a commercial air ambulance for transport to the hospital.

The injured sawyer was treated for 2nd degree burns on his lower arm and hand and was released from the hospital that same day.

2. The Story

It started as a normal day for an Interagency Hotshot Crew (IHC): a radio briefing from the Incident Management Team to their spike camp on expected weather and fire behavior for the day; overall strategy and tactics; and a message on safety. Next came another briefing with the Division Supervisor, further explaining the plan for that day.

The mission for Ruby Mountain IHC was to secure a slop-over from the day before that had burned into a green meadow area, leaving a spotty "dirty" burn.

This area needed to be secure before crews could begin pushing farther down the ridge, burning out the dozer line that had been prepped previously.



The area of the fire where the sawyers had been felling hazard trees.

As Day Heats Up, Chainsaws Start Dying

When they got to the slop-over area most of the crew began to "seek and destroy" spot fires out in the meadow, lining and securing them. Meanwhile, two saw teams worked along the fire's main edge, falling hazard trees to create a safe area for the crew to work in. The Crew Superintendent began scouting farther down the ridge, looking at the upcoming work, as well as pre-identifying safety zones, escape routes, and medevac/helispots if needed.

Work was progressing well. The saw teams finished their snagging in DIV J and began pushing into DIV I territory. As the day began to heat up, both saw teams were having issues with their chainsaws dying on them. The sawyers would have to step away from the hot areas in which they were working and allow their saws to cool down.

On Saw Team 1, after the faller would cut the hazard tree, the swamper would "low stump" it for him, giving the faller a break and providing a training opportunity for the less-experienced swamper. At one point, the swamper's saw died. The swamper carried the saw over to a green area and opened the cap to check the fuel. He noted at that time that the fuel was boiling inside the tank. He recapped it and observed that the fuel was still boiling.

"I was ticked that I had just got fuel sprayed all over me!"

Injured Sawyer



The hand tool location (in photo's left foreground) indicates the site where the fuel geyser occurred.

The swamper radioed the Saw Boss and asked if that was normal. The Saw Boss, not quite sure what the swamper meant or how he came to the conclusion that the fuel was boiling, answered: *"It's hot out here. You should let the saw cool down before continuing."*

Meanwhile, Saw Team 2 was working in a draw where smoke had pooled and hadn't yet cleared. After cutting for a few minutes, the sawyer would have to walk up out of the draw and get some fresh air and let his boots cool down. The swamper for Saw Team 2 was above him, staying clear of the felling area until needed.

This team's saw died again. The sawyer walked away from the hot area in which he was working to check the saw's fuel. He held the saw with his left hand and opened the cap with his right hand.

Fuel Cap Blows Off—Fuel Geysers Out of Tank

When he turned the cap it immediately blew off and fuel "geysered" from the tank, toward the sawyer's face. He quickly put his hand up to protect his face. He successfully protected his face with his gloved hand, but as a result, fuel sprayed up his arm and right side.

At the same time, his swamper, who was just up the hill from him, saw the sawyer walk out of the smoky area where he had been cutting. Assuming the sawyer was needing fuel, the swamper began heading his direction. He was able to observe the sawyer open the cap and witnessed the fuel spraying out of the tank.

The swamper watched as some of the fuel sprayed onto the sawyer while more shot up into the air and then fell back down onto the saw. When the fuel hit the saw it ignited and quickly engulfed the saw, the sawyer's arm, and his pack in flames. The sawyer quickly threw the saw away. He was unaware that he was on fire.

"I knew by the tone of his voice that something was wrong. He'd never said anything like that before."

Saw Boss

Swamper Tells Sawyer to Drop and Roll

The swamper got to him and told him to toss his pack down and to *"drop and roll!"* The sawyer moved to a greener, grassy area and began to roll and throw dirt on his arm, extinguishing the flames on and under his Nomex shirt. He then went back to his pack, picked up his radio and asked his Saw Boss to *"Bump down here real quick."*

Saw Boss Hustles to Scene

Sensing the urgency in the sawyer's voice, the Saw Boss hustled to the scene. When he got there, the sawyer was pacing and agitated, holding his arms out in front of him. He told the Saw Boss: *"I haven't looked at it yet. How bad is it?"*

The Saw Boss looked at the sawyer's arm.

He could see the burn marks and drooping skin.

He quickly called for the nearest crew EMT.

Still not sure what had just happened, the Saw Boss looked at the sawyer's arm. He could see the burn marks and drooping skin. The Saw Boss quickly called for the nearest crew EMT. He then sat the injured sawyer down, gave him a chew, and said: *"Uncle Dave* [the EMT] *is coming and everything's going to be alright."*

Saw Boss Becomes IC of this Incident Within an Incident

Next, the Saw Boss notified the crew's Assistant Superintendent and declared himself IC of this Incident Within an Incident. Then the Saw Boss gave his camera to the swamper and asked him to put the saw out, which was still on fire, and to start taking pictures of the scene.

As the crew EMT was in route to the scene, he called the other crew EMT who was carrying the first aid kit and asked him to meet him at the injury location. When the EMTs arrived on scene they first evaluated the area to make sure it was a safe to remain there to work on the injured sawyer. Deciding that it was, they began their assessment.

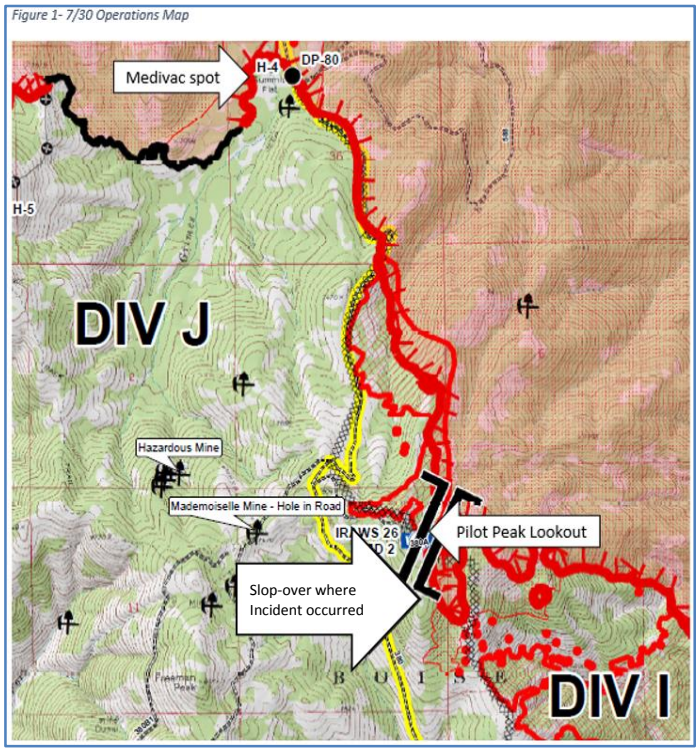
The Saw Boss filled out the "9 Line" while the EMTs concluded their assessment and began treatment of the injured sawyer by cooling his burn injury with their drinking water. It was determined that the injured sawyer would be able to make the 20-minute hike off the fireline under his own power to Pilot Peak to meet the Division Paramedic.

3. The Medevac

At 1420, the Division J Paramedic heard the Saw Boss read the "9 Line" over the Division's tactical frequency. The Division EMT who was with this Paramedic began copying down the medical report information on a "9 Line" form that he carried in his pocket.

The crew EMT and injured sawyer reported over the radio that they would be hiking to the Division Paramedic and EMT's location at Pilot Peak. While waiting for the injured sawyer to come to their location, the Divisional medical personnel began planning the medical evacuation.

Based on the information from the "9 Line," the Division Paramedic and EMT believed that the injured sawyer would need to be transported to a burn center.



Map of injury location, Helispot H-4, and Pilot Peak Lookout.

H-4, approximately a 20-minute drive from Pilot Peak, was the nearest helispot. The Division Paramedic called for the mini-ambulance, an off-road medical transport vehicle assigned to Division E, to rendezvous with them and the injured sawyer at Pilot Peak for transport to H-4.

As the injured sawyer and crew EMT were hiking toward the Division Paramedic's location they were met by another crew. This crew gave the crew EMT and the injured sawyer their ATV. This ATV was then used to transport the injured sawyer to the Paramedic much faster, cutting their travel time in half.



The mini-ambulance assigned to the Pioneer Fire, Division E.

"Using the time to pre-plan the medevac while we were waiting for the patient to come to us was very helpful. I'm glad I didn't wait for the mini-ambulance to arrive before starting transport to the helispot."

Line Paramedic

New, Updated 9 Line Filled Out and Transmitted to the ICP

The crew EMT and injured sawyer arrived at Pilot Peak and made contact with the Division medical personnel at 1440. The Division Paramedic began a patient assessment while the Division EMT, who transitioned to the medical incident IC, began filling out a new "9 Line".

The Division Paramedic continued cooling the sawyer's burn injuries with water and started an IV.

At 1454, the Division EMT walked away from the scene to use a mobile radio in a nearby vehicle to transmit the "9 Line" to the Incident Command Post (ICP). When the call came over the command frequency, the Radio Operator (RADO) at ICP immediately cleared the frequency for a medical emergency. At this time, all other activities in the communications tent ceased and the communications team readied themselves to receive the "9 Line".

"I believe that a thorough course on handling an Incident Within an Incident should be part of every firefighter's training, even at the FFT2 level."

Incident Communications Center Manager

Additional Assistance and Support Provided

Another RADO was assigned to assist the primary RADO by scribing the information from the report. In addition, the Incident Communications Center Manager (INCM) trainee stood close by to provide support to the RADO and to make contact with other resources needed for the medevac.

The Communications Unit Leader was out on the fire. Therefore, a Communications Technician who was in the tent notified the command and general staff that a medical emergency was ongoing.

At the beginning of the medical report to communications, an incorrect crew name was used but was corrected later in the report. The use of the crew name over the radio sparked a chain reaction of texts and social media posts that led to the injured firefighter receiving a text, even before he was transported off the incident, asking who on his crew had been hurt.

How Best Get Injured Sawyer to Definitive Care?

As the medical incident continued to unfold, the available command and general staff members came into the communications tent and listened quietly as the Medical Unit Leader (MEDL), who was now standing with the INCM in the communications tent, began to determine how best to get the injured sawyer to definitive care.

Although the Medical Incident IC had requested that an air ambulance be dispatched to H-4, the MEDL made the decision to use the pre-identified incident medevac ship to transport the injured sawyer from H-4 to the Pioneer Helibase (helibase) where it would rendezvous with the air ambulance.

The MEDL made this decision despite repeated requests from the Medical Incident IC to have the air ambulance go directly to H-4.

Decision to Use Incident Medevac Ship was Based on Several Factors

While the line medical personnel involved believed that having the air ambulance come directly to H-4 would reduce the overall transport time by 15-20 minutes, the MEDL believed that the incident medevac ship, which had been performing cargo missions on the fire, had the best chance of performing a successful and timely extraction of the injured sawyer from the incident.

This decision was also based on pre-planning that had been done for this type of incident and during the MEDL's previous experience. As the MEDL stated: *"I believe in doing things that have worked well for me in the past."*

The incident Safety Officers supported this decision due to their personal experience with a number of air ambulances having difficulties with locating helispots, communicating with ground resources, and pilot comfort levels with landing at remote helispots during direct medevacs from other wildfire incidents.

Review of Past Medevacs Involving Commercial Air Ambulances

Deer Park Wildfire FLA

"As the ensuing medical evacuation was in progress, a Life Flight helicopter was dispatched to the scene. Ground personnel were repeatedly unable to contact the helicopter. After landing at the medevac spot, the helicopter tipped backwards onto its tail boom, rendering the helicopter and medevac spot unusable for the medical evacuation."

Logging Unit Medivac RLS

"Several attempts were made by ground personnel to contact the Life Flight helicopter on Air to Ground and Air Guard frequencies assigned to the fire incident. Due to the lack of communications with Life Flight, the Life Flight ship was confused about the location of the Medevac site even though they had a lat/long. Life Flight flew directly over the medevac site and attempted to land at the drop point a mile and a half down the mountain."

Incident's Helitack Crew Prepares for Medevac Mission

During this time, members of the Helitack Crew of the Pioneer Fire's designated medevac helicopter were listening to the radio traffic over command. The helicopter was currently performing a long line mission on the fire. Its crew immediately began to make preparations for performing the medevac.

The medevac helicopter and crew are a designated Short-Haul unit. As such, the crew and pilot are trained in performing emergency medical evacuations and are ready and available to put their training into action. The crew at the helibase prepared the

doors (the doors were off for the long line mission) for installation and gathered up medical gear to be put onboard the ship.

Medevac Ship Quickly Reconfigures for Mission

At 1459, the INCM contacted the helibase requesting that the medevac ship return to helibase to prepare for the medevac mission. Because this same crew had performed a medevac off the Pioneer Fire a couple days earlier, this mission's directions from the INCM were simple: *Head to H-4 and pick-up the injured sawyer*.

> ***"Remember what we did the other day? We are doing that again."***
>
> **Communications to Helibase from the Incident Communications Center Manager**

When asked where the medevac ship would rendezvous with the air ambulance, the INCM replied: *"Remember what we did the other day? We are doing that again."*

The helicopter landed at helibase at 1511.The crew quickly removed the long line, installed the doors, and loaded the medical gear. Six minutes later the helicopter was off the helibase enroute to H-4 with the Helicopter Manager and an EMT.

> ***"Good training fell into place from the crew, to the medics, to the IMT."***
>
> **Crew Superintendent**

All Relevant Info Relayed to Air Ambulance

Back at the communications tent, following instructions given during the team in-briefing with the Forest Supervisor to follow the local Interagency Dispatch Center (Dispatch Center) incident medical response protocols and to coordinate all medical responses with the Dispatch Center, the MEDL contacted the Dispatch Center and requested an air ambulance, as well as a back-up ground ambulance, to come to the Pioneer Helibase.

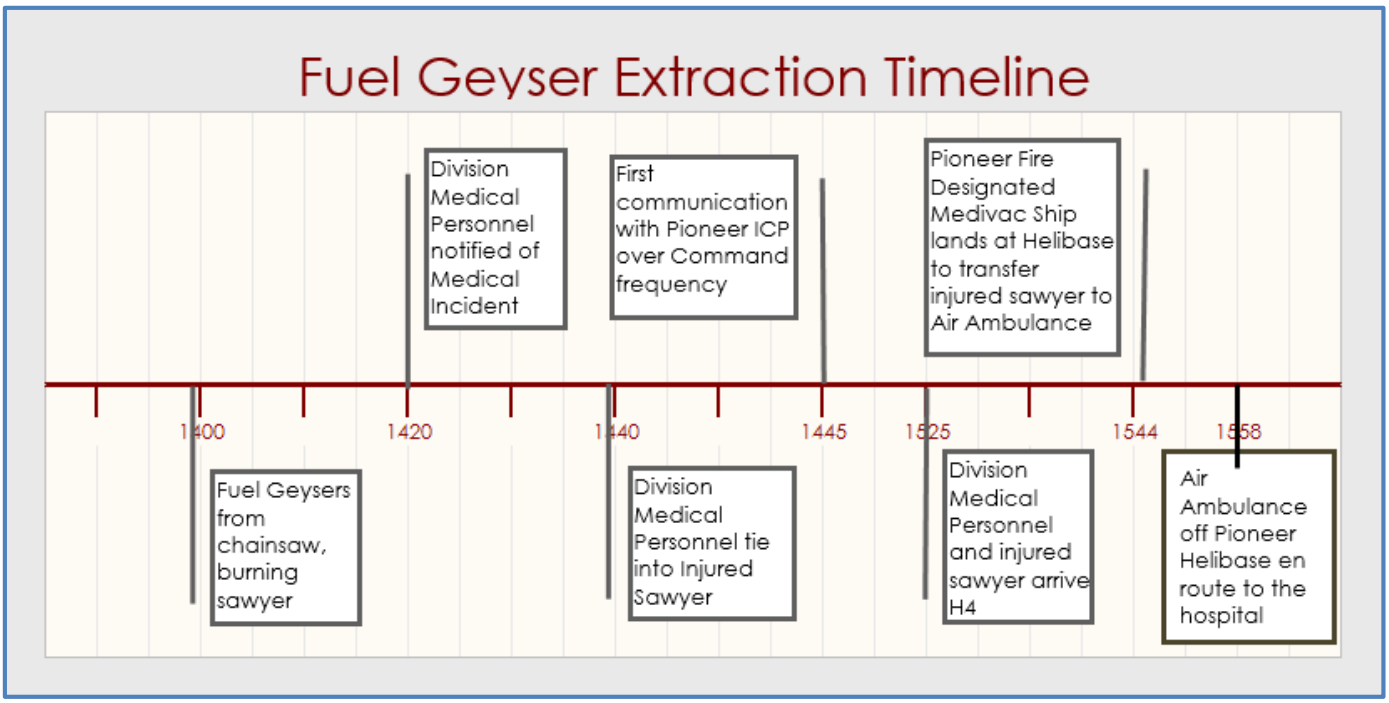
During this contact, the Dispatch Center took basic information on the medical incident and the requested location for the air ambulance to land. This information was then relayed to the State Communications Center and the State EMS Dispatch, who is responsible for dispatching the closest air and ground ambulances.

The Dispatch Center also confirmed the appropriate radio frequencies to be used. Once all the information was passed onto the State Communications Center and the frequencies were confirmed, the Dispatch Center called the Pioneer Fire back to relay this information and inform them that the anticipated arrival time for the air ambulance to helibase was approximately 1540.

The Dispatch Center also asked for any updates on the injured sawyer's condition to be passed on to the medical responders.

The working relationship that the Dispatch Center has with the local EMS providers and the response plan helped to ensure a successful transfer of the injured sawyer from the field to definitive care using the most effective method.

> ***"This opened my eyes to transport times. We were not in a remote area and it took an hour and 20 minutes. How long would it have taken if we had been remote?"***
>
> **Crew Superintendent**



Division Medical Personnel and Injured Sawyer Meets Mini-Ambulance at Pilot Peak
Around this same time, the Division medical personnel made the decision to transport the injured sawyer to H-4 with their SUV rather than waiting for the mini-ambulance to arrive at Pilot Peak. They began this trip at 1508 and met the mini-ambulance on the road to H-4. The paramedic on the mini-ambulance had pain medication that was administered to the injured sawyer along with an anti-nausea medication. This substantially reduced the injured sawyer's pain level and the transport to H-4 continued.

While completing the transport to H-4, the Division Paramedic contacted Pioneer Communications and reported that the injured sawyer was stable, they would be arriving at H-4 soon, and that the command frequency could resume normal traffic. The RADO opened up command and the emergency medical traffic was terminated at 1513.

Medevac Helicopter Flies Injured Sawyer to Helibase to Rendezvous with Air Ambulance
At 1525, both the injured sawyer and the incident medevac helicopter arrived at H-4. Prior to loading into the helicopter, the injured sawyer's burn injury was wrapped with a dry sterile dressing.

Coincidentally, some of the crew members from the incident medevac helicopter had been working at H-4 and were able to serve as the ground contact and assist with loading the injured sawyer.

The medevac helicopter lifted off from H-4 with the injured sawyer, Division Paramedic, and helicopter crew at 1535 enroute to the helibase to rendezvous with the air ambulance.

MEDL Secures Golf Cart to Help Shuttle Injured Sawyer from Medivac Helicopter to Air Ambulance
Meanwhile, the MEDL had left the communications tent for the helibase (a two-minute walk across base camp) to assist with coordinating the transfer of the injured sawyer to the air ambulance. The helibase for the Pioneer Fire is particularly large with a substantial distance between the incident medevac helicopter's assigned landing pad and where there was space available for the air ambulance to land. To remedy this, the MEDL secured a golf cart assigned to the helibase to be used to transfer the injured sawyer.

Injured Sawyer's Care Transferred to Air Ambulance Flight Nurse

At 1544, the incident medevac helicopter landed at helibase and the air ambulance arrived immediately thereafter. The injured sawyer was transported via the golf cart to the air ambulance and the transfer of patient care to the Flight Nurse was completed.

> ***"The medical incident ran smooth. I was very impressed. The crew was dialed, they followed their training and practice. This kept me calm."***
>
> **Injured Sawyer**

The Flight Nurse was supplied with the Red Book burn criteria for admission to a burn unit. The Flight Nurse completed a new patient assessment and believed that, based on the burn criteria, immediate transport to a burn center was not warranted. At 1558, the air ambulance lifted off from the helibase enroute to the hospital.

Hospital Liaisons Identified – for both the Local Hospital and Salt Lake City Burn Center

Once they received the call about the medical incident from the Pioneer Fire, the Dispatch Center, working with Forest leadership, identified a Hospital Liaison to assist the injured sawyer when he arrived at the hospital.

The local National Forest's Emergency Medical Response Plan directs the responsible agency administrator to:

> ***"…assign a person to act as liaison with the hospital. This person should perform this important function full time through the first critical days. Avoid assigning someone with collateral duties that would interfere with the duties of Hospital Liaison. Assign a person to handle comp/claims paperwork with ASC."***

Because the injured sawyer was a Bureau of Land Management employee working on a fire under U.S. Forest Service administrative authority, a representative from both the BLM and Forest Service were sent to the hospital to provide support for the injured sawyer.

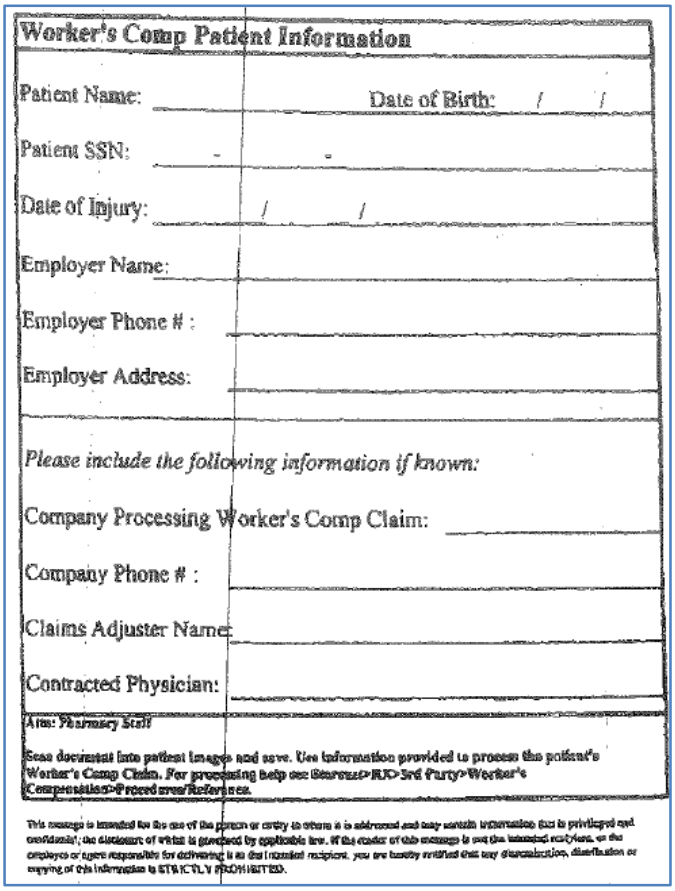
In addition, because it was initially unknown if the injured sawyer would be sent to the local hospital or to the University of Utah Health Care, Burn Center in Salt Lake City, a Hospital Liaison in Salt Lake was also arranged and placed on standby. Once it was confirmed that the injured sawyer was going to the local hospital, this liaison was cancelled.

Good Sign: the Injured Sawyer and Hospital Liaison Know Each Other

The Hospital Liaison arrived at the local hospital and went to the registration desk at the Emergency Room (ER). The liaison explained who she was and what her duties were. Initially, the registration personnel at the ER were confused. But after the Hospital Liaison explained her role, this confusion was cleared up and she was allowed access to the injured sawyer.

> ***"It was really cool, they had a liaison at the hospital to handle all the paperwork for me and my Superintendent."***
>
> **Injured Sawyer**

As luck would have it, the injured sawyer and the Hospital Liaison knew each other. This familiarity helped to lower the injured sawyer's stress level. The liaison was able to get the paperwork (CA-1 and CA-16) process started and called the Crew Superintendent to let him know she was at the hospital. In addition, the injured sawyer's family lives in the local area and was starting to arrive at the hospital.



Worker's Comp Patient Information

Patient Name: _____ Date of Birth: / /

Patient SSN: - -

Date of Injury: / /

Employer Name:

Employer Phone # :

Employer Address:

Please include the following information if known:

Company Processing Worker's Comp Claim:

Company Phone # :

Claims Adjuster Name:

Contracted Physician:

Attn: Pharmacy Staff

Scan document into patient images and save. Use information provided to process the patient's Worker's Comp Claim. For processing help see: Scanning>RIP>3rd Party>Worker's Compensation>Procedures/References.

This message is intended for the use of the person or entity to whom it is addressed and may contain information that is privileged and confidential; the disclosure of which is governed by applicable law. If the reader of this message is not the intended recipient, or the employee or agent responsible for delivering it to the intended recipient, you are hereby notified that any dissemination, distribution or copying of this information is STRICTLY PROHIBITED.

The Hospital Liaison was also able to help explain what was going on in regards to the treatment of the injured sawyer and how the OWCP process worked. She stayed with the injured sawyer at the hospital and with his family until he was released.

Hospital Liaison Travels to Pharmacy to Help and also Provides an Array of Follow-Up Services

Once the injured sawyer was released from the hospital, the Hospital Liaison drove to the pharmacy to assist in filling out his prescription medications. At the pharmacy they also filled out the pharmacy's "Worker's Comp Patient Information" claim form. By filling out this form, the injured sawyer did not have to use his insurance or pay for the medications. The pharmacy was able to bill OWCP directly.

After the injured sawyer was released from the hospital and his prescriptions were filled, the Hospital Liaison's job was still not done. She checked in with the injured sawyer daily making sure he had transportation to and from doctor's appointments, additional bandages, and other medical supplies. She also kept him informed as to the progress of his OWCP claim and was always available if he had any questions that needed answered.

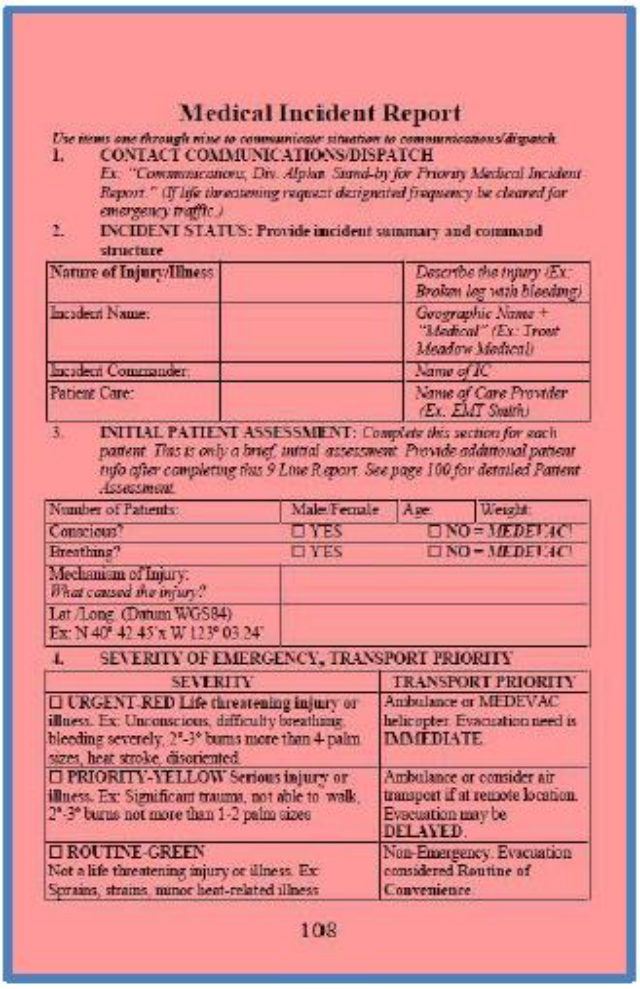
Follow-Up Doctor Appointments

For several days after being released from the hospital, the injured sawyer had follow-up doctor appointments to continue treatment and monitoring of his burn injuries.

Five days after the injury occurred, he had a follow up appointment via "Skype" with the Burn Center in Salt Lake City. A week later another follow-up appointment with the burn center occurred, once again via "Skype", to determine if skin grafts were needed and to give the injured sawyer an anticipated time line for when he could return to work.

4. What was Learned?

- ✓ A fuel geyser is not the "little hiss and spray" that we are used to. All the hotshot crews know and talk about the geysers, but this is more than we understand it to be. Fuel Geysering really does happen.
- ✓ Even if you get away from the fire, the fuel from a geyser may still ignite. Performing other saw maintenance before fueling, such as cleaning the air filter or sharping the chain, is one way to allow more time for the saw to cool.
- ✓ Training on filling out and reading the "9 Line" over the radio should be a priority for every crew. Think about the person on the other end who is responsible for writing everything down. Talk slowly and take breaks between sections.
- ✓ Even if you declare an emergency and request to clear radio traffic, other resources may still attempt to use the radio frequency. Many times, other resources are unable to hear the "Clear Traffic" or the radio traffic following. A second "Clear Traffic" may be necessary.



Medical Incident Report

Use items one through nine to communicate situation to communications/dispatch.

1. CONTACT COMMUNICATIONS/DISPATCH
Ex: "Communications, Div. Alpha. Stand-by for Priority Medical Incident Report." (If life threatening request designated frequency be cleared for emergency traffic.)
2. INCIDENT STATUS: Provide incident summary and command structure

Nature of Injury/Illness:		*Describe the injury (Ex: Broken leg with bleeding)*
Incident Name:		*Geographic Name + "Medical" (Ex: Trout Meadow Medical)*
Incident Commander:		*Name of IC*
Patient Care:		*Name of Care Provider (Ex. EMT Smith)*

3. INITIAL PATIENT ASSESSMENT: *Complete this section for each patient. This is only a brief, initial assessment. Provide additional patient info after completing this 9 Line Report. See page 106 for detailed Patient Assessment.*

Number of Patients:	Male/Female	Age:	Weight:
Conscious?	☐ YES	☐ NO = *MEDEVAC!*	
Breathing?	☐ YES	☐ NO = *MEDEVAC!*	
Mechanism of Injury: *What caused the injury?*			
Lat /Long (Datum WGS84) Ex: N 40° 42 45 x W 123° 03 24'			

4. SEVERITY OF EMERGENCY, TRANSPORT PRIORITY

SEVERITY	TRANSPORT PRIORITY
☐ URGENT-RED Life threatening injury or illness. Ex: Unconscious, difficulty breathing, bleeding severely, 2°-3° burns more than 4 palm sizes, heat stroke, disoriented.	Ambulance or MEDEVAC helicopter. Evacuation need is IMMEDIATE.
☐ PRIORITY-YELLOW Serious injury or illness. Ex: Significant trauma, not able to walk, 2°-3° burns not more than 1-2 palm sizes	Ambulance or consider air transport if at remote location. Evacuation may be DELAYED.
☐ ROUTINE-GREEN Not a life threatening injury or illness. Ex: Sprains, strains, minor heat-related illness	Non-Emergency. Evacuation considered Routine of Convenience

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The Medical Incident Report "9 Line" is located inside the Incident Response Pocket Guide (IRPG) on page 108.

- ✓ Once the medical emergency is stabilized and a solid plan is in place, opening the frequency back up for normal radio operations is OK—even though the incident may still be ongoing. This allows necessary operational traffic to resume quicker, without affecting the emergency traffic. The frequency can always be cleared again if needed.
- ✓ Do not mention the names of the crew or other resources involved in a medical incident over the radio. During this incident, an incorrect crew named was relayed over the radio during the medical report but was later corrected. Due to the crew name being verbalized over the radio, a chain reaction of text and social media posts began. This led to the injured sawyer receiving a text asking who on the crew was hurt, even before he was transported off the incident.

- ✓ "We read the Lessons Learned Center's postings, but before this, I really didn't pay much attention to them. I will now take them more seriously."
- ✓ We know there is an issue. And we know there are steps you should take to mitigate that issue. But after days—or even weeks—of doing the same thing, it's easy to get complacent and revert back to the way you've always done things. Sawyers need to take a tactical pause when they need to refuel or if they are having issues with their saw running poorly. Stop, Think, and Then Act.
- ✓ Having paramedics actually out on the line with the crews expedites getting injured personnel the care they need.

Reviews of Past Fuel Geysers

Homestead Fire Burn Injury
June 16, 2016 (2 pages). Burn injuries sustained when fuel erupted from a dolmar and ignited during refueling operations on initial attack.

Willow Creek Burn Injury
2015 (22 pages). A sawyer receives serious burn injuries when fuel sprays out of the chainsaw's gas tank and ignites.

Chainsaw Fuel Mishap - Rock Ridge Burn Injury
2014 (1 page). A county firefighter received severe burns to the forearms and legs while operating a chainsaw around the perimeter of the fire. The chainsaw was found at the incident site with the fuel cap off and the tank empty. Witnesses saw the fire ignite at ground level and then engulf the firefighter.

"We read the Lessons Learned Center's postings, but before this, I really didn't pay much attention to them. I will now take them more seriously."

- ✓ Going to an Emergency Room is often more efficient than going to an urgent care facility, especially when the patient is not from the area. At the ER, patients can often get the complete care they need for minor injuries and illness in one visit, including getting prescriptions.
- ✓ Even though the forms from OWCP are the same for all agencies, the databases and reporting processes are different for each agency. The injured sawyer was a BLM employee working on a fire under Forest Service administrative authority. Having a liaison from both agencies helped navigate these databases and processes.
- ✓ Pre-assigning a local Hospital Liaison to an ongoing fire incident allows for continuity in services by providing local knowledge of hospitals and pharmacies in the area, ensuring that the patient has faster access to proper treatment as well as prescription filling.
- ✓ Establishing pre-season relationships with local hospital ERs could help clear up the hospital's confusion regarding the role of the Hospital Liaison position.
- ✓ Crews and individuals do not appreciate being asked multiple times for interviews or being repeatedly asked to tell their story of what happened. To reduce this concern, determine early on the level of review that will be conducted for the incident and request any interviews/AARs to be postponed until the Review Team can participate.

5. What Helped Make Things Go 'Right' During This Medical Incident?

- ✓ The "9 Line" was used by all parties to accurately transmit information as the medical incident evolved.
- ✓ Pre-planning for a medical incident had been performed at all levels (crew, line medics, communications, medical unit, and incident command).
- ✓ Having three EMTs on the crew with diverse backgrounds and experience.
- ✓ At all levels, identifying a scribe for the "9 Line" allowed others to focus on the task at hand.
- ✓ Using learning gained from previous experiences.
- ✓ The crew kept all non-essential personnel away from the scene and the EMTs ensured that the scene was clear of overhead hazards.
- ✓ Letting trained personnel—no matter their job title or position—handle coordination of the response to the incident.
- ✓ The local National Forest had an established Hospital Liaison and training program in place and provided an agency representative to assist the injured sawyer and his family with paperwork and OWCP issues.
- ✓ Assigning a liaison from both the Forest Service and the BLM allowed for both agencies' processes to be completed.

> ***"We can only control what we can control. Good, realistic training all the way up the line and with everyone involved allowed for a fast evac and a good outcome."***
>
> **Crew Superintendent**

6. Employee Concerns

Throughout this Lessons Learned Review, the Review Team heard concerns surrounding the issue of fuel geysering from many fire personnel, including: firefighters, IMT members, and line officers.

The following quotes are representative examples of what the Review Team heard:

> ***"It makes me mad that one of our guys gets hurt from an identified safety issue that has not been fixed."***
>
> **Firefighter**

> ***"As an agency we identify problems but we are afraid to stick our necks out—whether it be short-haul, low angle rescue, or chainsaws."***
>
> **Firefighter**

"We identify a problem but we don't fix it. As a crew supervisor, I am part of this problem."

Firefighter

"We know there is a problem but we keep putting these chainsaws in these kid's hands and telling them to go to work."

Forest Leader

Most Current Information Regarding a Possible "Fix" for Fuel Geysering

Additionally, there is a widespread belief among wildland fire personnel that a "fix" has been developed by agency engineers that will eliminate fuel geysering. The paragraphs below are the most current information about the status of this "fix":

"The National Technology & Development Center in San Dimas is currently working on ways to prevent fuel "geysering" from occurring in chainsaws and other gasoline powered equipment and containers. A promising engineering solution for Stihl chainsaws with the tool-less fuel cap was replacing the current 0-ring on the cap with a larger 0-ring and the addition of a fiber rubber gasket. This modification would slow the release of pressure in the fuel tank when the tank is opened. Originally, a limited number of tests were performed with the modified fuel cap. The U.S. Forest Service provided details of this testing to the Stihl Corporation to help validate the tests, validate the potential solution, and also to assist in identifying other unfavorable unintended consequences. In further testing by San Dimas, results showed that the modified cap does NOT appear to provide a robust solution as hoped. More tests are ongoing, including field-testing taking place on the Pioneer Fire where the "geysering" issue was last experienced.

Many people are concerned that the issue is coming from the ethanol blended fuel we are now using. Ethanol or any oxygenates blended with fuel does cause chainsaws to run at a higher temperature. Ethanol is more volatile than gasoline, but the overall volatility of the blend is regulated by month and location. The most significant change is between winter and summer. Several tests have indicated that using "winter" blend gasoline in summer-like conditions can increase the likelihood of fuel geysering. The "winter" blends are designed to work in cold conditions and thus are more volatile. This volatility contributes to the build-up of pressure in the fuel tank. Unlike water, gasoline does not have a single boiling (or distillation) point because it is a blend of different hydrocarbons. These series of boiling points are described by a distillation curve. A change in elevation shifts this curve down and thus increases the risk of fuel geysering. For example, summer fuel purchased in Grand Junction, Colorado (at 4,600 feet) that in laboratory testing would produce a "geyser" when the fuel temperature reaches 134 degrees would most likely "geyser" when the fuel temperature is at 119 degrees at 8000 ft."

Currently, the best "fix" we have are the mitigation techniques that are described in this year's

"2016 WFSTAR Fireline Fuel Safety Video"



When you need to open the cap on your saw:

1. Walk away from any heat source.
2. Set the saw on the ground and allow a 5-minute cool-down period.
3. With the top of the saw pointed toward you, tip the saw away from you and open the cap SLOWLY!!!
4. If you have a rag, plastic sack, or a glove that you can put over the cap when opening, do so. This will help direct any fuel spray away from you.

REMEMBER: STOP, THINK, (TALK) AND THEN ACT

7. Lessons Learned Review Team

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